# Outdoors enzyme statistical analysis

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# Load packages

```
library(tidyverse)
library(lme4)
library(lmerTest)
library(emmeans)

dir.create("Outdoors enzyme results", showWarnings = FALSE)
```

# Import data

```
df <- read.csv("all_enzyme_data_outdoors.csv")
```

# Prepare variables

```
df$Species <- as.factor(df$Species)

df$Treatment <- as.factor(df$Treatment)

df$`Plant.part` <- as.factor(df$`Plant.part`)

df$Enzyme.function <- as.factor(df$Enzyme.function)

df$Enzyme <- as.factor(df$Enzyme)
```

# Overall model

```
model <- lmer(
  Log2.Fold.Change ~
    Enzyme.function *
    Treatment *
    Species *
    Plant.part +
    Experiment +
    (1 | Enzyme.function:Enzyme),
  data = df
)
```

# ANOVA

```
anova_results <- anova(model)

anova_results
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##                                          Sum Sq Mean Sq NumDF   DenDF
## Enzyme.function                           4.014   1.004     4  13.406
## Treatment                                19.441  19.441     1 226.525
## Species                                  13.610  13.610     1 226.525
## Plant.part                                0.195   0.195     1 226.525
## Experiment                               33.403  33.403     1 231.443
## Enzyme.function:Treatment                 1.043   0.261     4 226.525
## Enzyme.function:Species                  14.764   3.691     4 226.525
## Treatment:Species                         7.427   7.427     1 226.525
## Enzyme.function:Plant.part               12.206   3.051     4 226.525
## Treatment:Plant.part                      0.438   0.438     1 226.525
## Species:Plant.part                       20.199  20.199     1 226.525
## Enzyme.function:Treatment:Species         0.821   0.205     4 226.525
## Enzyme.function:Treatment:Plant.part      4.346   1.086     4 226.525
## Enzyme.function:Species:Plant.part        4.303   1.076     4 226.525
## Treatment:Species:Plant.part              5.721   5.721     1 226.525
## Enzyme.function:Treatment:Species:Plant.part  0.855   0.214     4 226.525
##                                          F value    Pr(>F)
## Enzyme.function                           0.5607 0.6951738
## Treatment                                10.8618 0.0011394 **
## Species                                   7.6038 0.0062997 **
## Plant.part                                0.1090 0.7415667
## Experiment                               18.6622 2.317e-05 ***
## Enzyme.function:Treatment                 0.1457 0.9647690
## Enzyme.function:Species                   2.0622 0.0866571 .
## Treatment:Species                         4.1495 0.0428086 *
## Enzyme.function:Plant.part                1.7049 0.1498001
## Treatment:Plant.part                      0.2445 0.6214563
## Species:Plant.part                       11.2854 0.0009165 ***
## Enzyme.function:Treatment:Species         0.1147 0.9772544
## Enzyme.function:Treatment:Plant.part      0.6070 0.6580162
## Enzyme.function:Species:Plant.part        0.6011 0.6622410
## Treatment:Species:Plant.part              3.1961 0.0751522 .
## Enzyme.function:Treatment:Species:Plant.part  0.1195 0.9754674
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
write.csv(
    anova_results,
    "Outdoors enzyme results/ANOVA_Table.csv"
)
```

# Estimated marginal means

# Treatments

```
emm_2 <- emmeans(model,  ~ Treatment | Enzyme.function * Species * Plant.part)

emm_pairs_2 <- pairs(emm_2, adjust = "BH")
emm_pairs_2
```

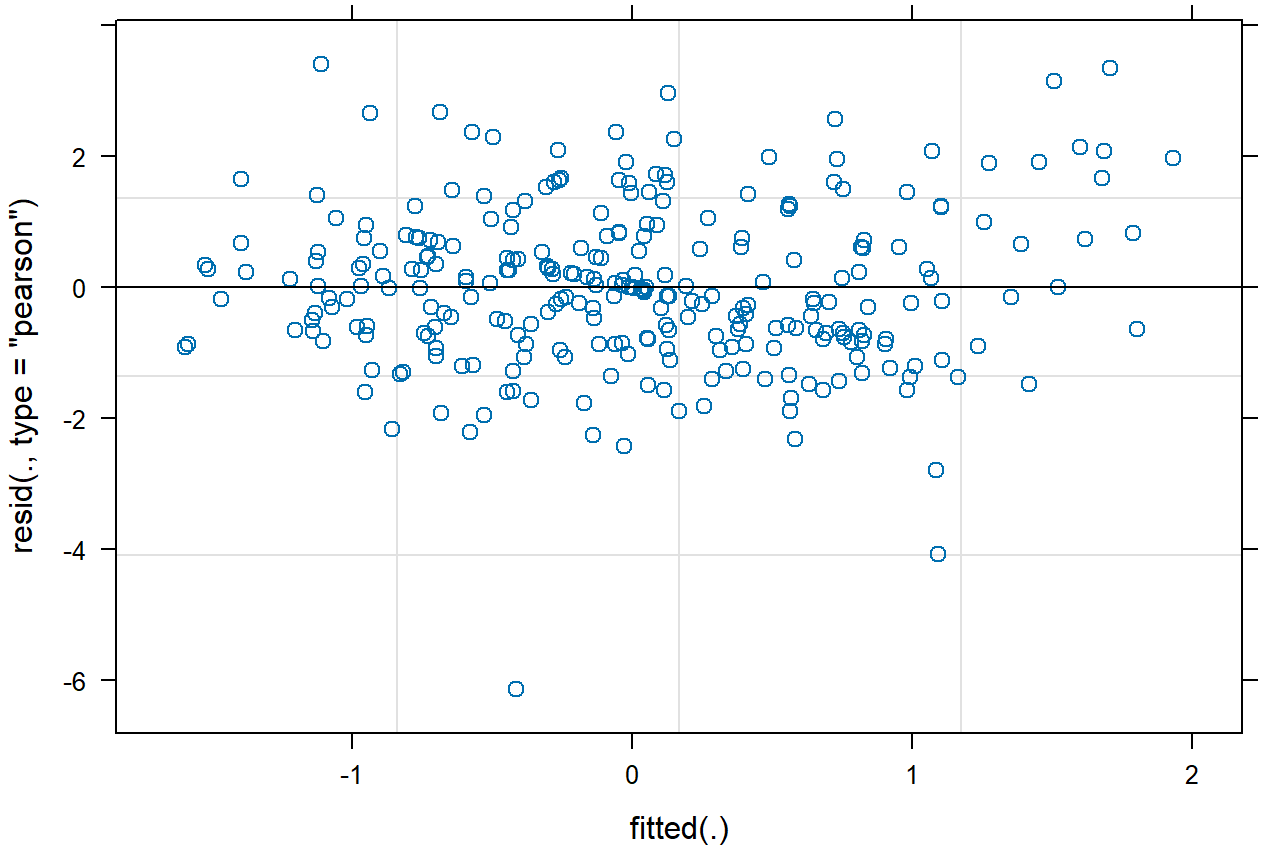
```
## Enzyme.function = Antioxidant, Species = D. flexuosa, Plant.part = Root:
##  contrast                    estimate    SE  df t.ratio p.value
##  R freezing - (R+S freezing)  -0.2744 0.546 226  -0.502  0.6158
##
## Enzyme.function = Carbon redistribution, Species = D. flexuosa, Plant.part = Root:
##  contrast                    estimate    SE  df t.ratio p.value
##  R freezing - (R+S freezing)   0.1659 0.846 226   0.196  0.8447
##
## Enzyme.function = Carbon storage, Species = D. flexuosa, Plant.part = Root:
##  contrast                    estimate    SE  df t.ratio p.value
##  R freezing - (R+S freezing)  -0.1128 0.772 226  -0.146  0.8840
##
## Enzyme.function = Glycolysis, Species = D. flexuosa, Plant.part = Root:
##  contrast                    estimate    SE  df t.ratio p.value
##  R freezing - (R+S freezing)  -0.2483 0.772 226  -0.322  0.7481
##
## Enzyme.function = Sucrolytic activity, Species = D. flexuosa, Plant.part = Root:
##  contrast                    estimate    SE  df t.ratio p.value
##  R freezing - (R+S freezing)   0.4407 0.772 226   0.571  0.5689
##
## Enzyme.function = Antioxidant, Species = E. vaginatum, Plant.part = Root:
##  contrast                    estimate    SE  df t.ratio p.value
##  R freezing - (R+S freezing)   1.5279 0.546 226   2.797  0.0056
##
## Enzyme.function = Carbon redistribution, Species = E. vaginatum, Plant.part = Root:
##  contrast                    estimate    SE  df t.ratio p.value
##  R freezing - (R+S freezing)   1.3284 0.846 226   1.570  0.1178
##
## Enzyme.function = Carbon storage, Species = E. vaginatum, Plant.part = Root:
##  contrast                    estimate    SE  df t.ratio p.value
##  R freezing - (R+S freezing)   0.9787 0.772 226   1.267  0.2064
##
## Enzyme.function = Glycolysis, Species = E. vaginatum, Plant.part = Root:
##  contrast                    estimate    SE  df t.ratio p.value
##  R freezing - (R+S freezing)   0.9820 0.772 226   1.271  0.2049
##
## Enzyme.function = Sucrolytic activity, Species = E. vaginatum, Plant.part = Root:
##  contrast                    estimate    SE  df t.ratio p.value
##  R freezing - (R+S freezing)   1.5589 0.772 226   2.018  0.0448
##
## Enzyme.function = Antioxidant, Species = D. flexuosa, Plant.part = Shoot:
##  contrast                    estimate    SE  df t.ratio p.value
##  R freezing - (R+S freezing)   0.2373 0.546 226   0.435  0.6643
##
## Enzyme.function = Carbon redistribution, Species = D. flexuosa, Plant.part = Shoot:
##  contrast                    estimate    SE  df t.ratio p.value
##  R freezing - (R+S freezing)   0.0899 0.846 226   0.106  0.9155
##
## Enzyme.function = Carbon storage, Species = D. flexuosa, Plant.part = Shoot:
##  contrast                    estimate    SE  df t.ratio p.value
##  R freezing - (R+S freezing)   0.5893 0.772 226   0.763  0.4463
##
```

```
## Enzyme.function = Glycolysis, Species = D. flexuosa, Plant.part = Shoot:
##  contrast                    estimate    SE  df t.ratio p.value
##  R freezing - (R+S freezing)   0.7678 0.772 226   0.994  0.3213
##
## Enzyme.function = Sucrolytic activity, Species = D. flexuosa, Plant.part = Shoot:
##  contrast                    estimate    SE  df t.ratio p.value
##  R freezing - (R+S freezing)   0.4524 0.772 226   0.586  0.5587
##
## Enzyme.function = Antioxidant, Species = E. vaginatum, Plant.part = Shoot:
##  contrast                    estimate    SE  df t.ratio p.value
##  R freezing - (R+S freezing)   0.2448 0.546 226   0.448  0.6545
##
## Enzyme.function = Carbon redistribution, Species = E. vaginatum, Plant.part = Shoot:
##  contrast                    estimate    SE  df t.ratio p.value
##  R freezing - (R+S freezing)   0.1750 0.846 226   0.207  0.8363
##
## Enzyme.function = Carbon storage, Species = E. vaginatum, Plant.part = Shoot:
##  contrast                    estimate    SE  df t.ratio p.value
##  R freezing - (R+S freezing)   0.5034 0.772 226   0.652  0.5152
##
## Enzyme.function = Glycolysis, Species = E. vaginatum, Plant.part = Shoot:
##  contrast                    estimate    SE  df t.ratio p.value
##  R freezing - (R+S freezing)   1.4623 0.772 226   1.893  0.0596
##
## Enzyme.function = Sucrolytic activity, Species = E. vaginatum, Plant.part = Shoot:
##  contrast                    estimate    SE  df t.ratio p.value
##  R freezing - (R+S freezing)   0.1686 0.772 226   0.218  0.8274
##
## Results are averaged over the levels of: Experiment
## Degrees-of-freedom method: kenward-roger
```

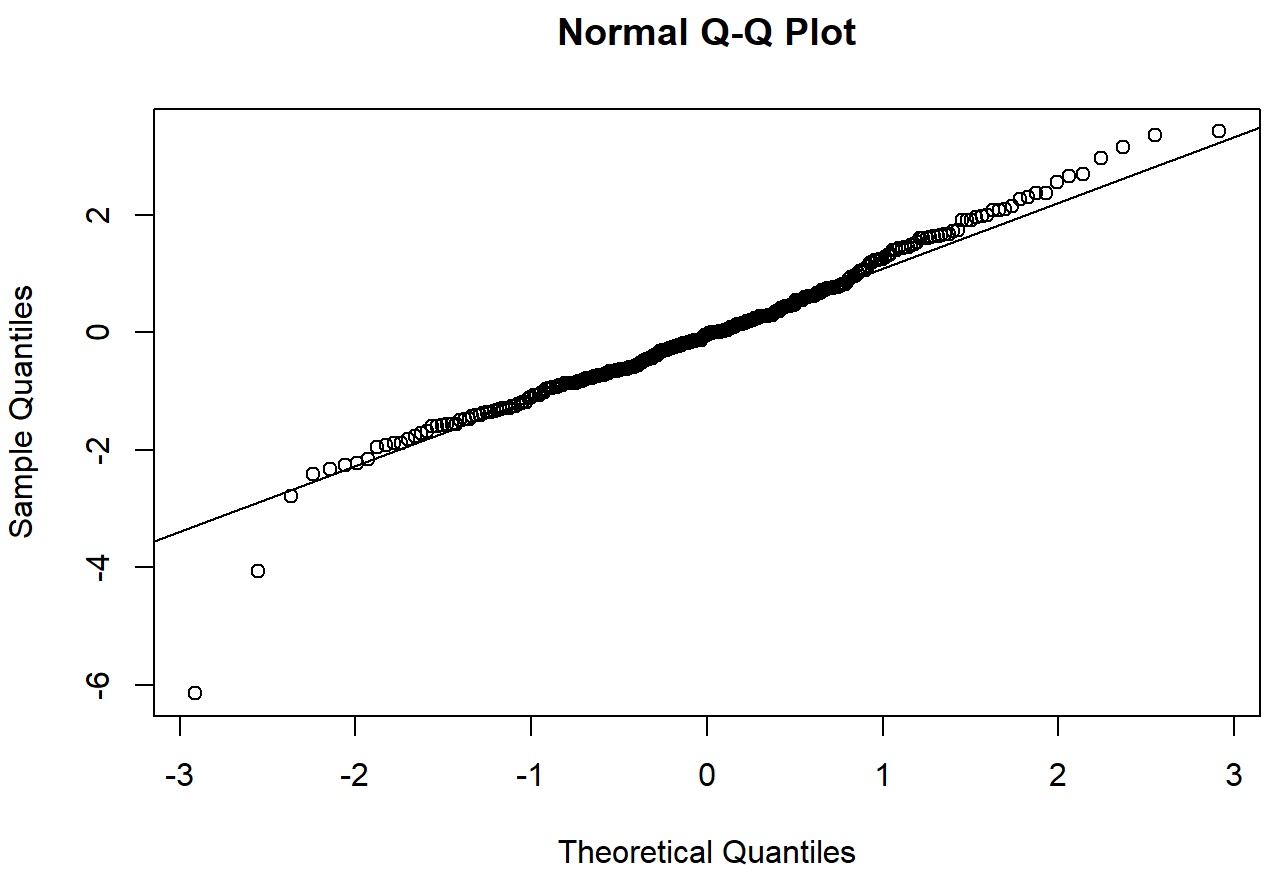
```
write.csv(emm_pairs_2,
          "Outdoors enzyme results/Treatments_Pairwise_Comparisons.csv",
          row.names = FALSE)
```

# Diagnostics

```
plot(model)
```




```
qqnorm(residuals(model))
qqline(residuals(model))
```


Normal Q-Q Plot
Sample Quantiles
Theoretical Quantiles
2
0
-2
-4
-6
-3
-2
-1
0
1
2
3